

Linear Circuit Analysis

Semester Project Report

AA Battery Voltage Tester

Constant-Current Load with ADC Measurement and I2C LCD Display



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1. Objective

This project was built as the final semester project of the Linear Circuit Analysis course. The goal was to apply the theoretical concepts covered throughout the semester, such as op-amp circuits, transistor biasing, voltage dividers, and ADC-based measurement, in a working hardware design that solves a real-world problem.

The objective of this project is to design and build an AA battery voltage tester that accurately measures the health of a 1.5V AA cell under a realistic load condition. Rather than simply measuring the open-circuit voltage of a battery, which can be misleading, the circuit applies a constant load current of approximately 116 mA to the battery while simultaneously reading its terminal voltage through a microcontroller. The result is displayed on a 16x2 LCD screen as both a voltage reading and a health status label, giving the user an honest assessment of whether the battery is still useful.

2. Problem Statement

A battery that appears fully charged at rest can collapse in voltage the moment it is made to supply real current. This is the fundamental limitation of open-circuit voltage testing. A fresh AA cell reads approximately 1.5V with nothing connected to it, but an old or partially discharged cell can also show a voltage close to that when unloaded. The difference only becomes apparent when the battery is asked to do actual work.

This project addresses that problem by designing a circuit that does the following:

- **Applies a fixed load:** The circuit draws a consistent 116 mA from the battery during the measurement. This is comparable to the drain of a TV remote control or a small sensor device, making the test realistic.
- **Measures terminal voltage under load:** The Arduino ADC reads the battery voltage while the load is active, not before or after, so the reading reflects how the battery performs under use.
- **Displays results clearly:** The LCD shows the measured voltage and a plain-language label such as GOOD, OK, LOW, or DEAD, along with a 16-block graphical health bar, so no interpretation is needed by the user.

The following specific problems were identified and addressed in the design:

- Open-circuit readings are unreliable for determining usable battery life, so a constant-current load circuit was required.

- The load current needed to be stable and well-defined, not dependent on battery voltage, so a closed-loop op-amp control circuit was used instead of a simple resistor.
- The ADC measurement path needed to be fully isolated from the load path to avoid reading a distorted voltage affected by resistor drops in the load circuit.
- The display needs to be refreshed regularly and shows information in a format that requires no technical knowledge to interpret.

3. Components and Their Roles

The following table lists every component used in the circuit along with its value and purpose.

Table 1: List of components

Component	Value	Role in the Circuit
R1	10 k Ω	Upper resistor of the reference voltage divider. Sets the 0.455 V reference for the op-amp.
R2	1 k Ω	Lower resistor of the reference voltage divider. Works with R1 to produce the 0.455 V reference.
R3	1 k Ω	Base current limiting resistor between the op-amp output and the transistor base. Protects both components and adds loop stability.
R4	100 k Ω	ADC protection resistor. Connects battery positive to Arduino pin A0. Limits current if a voltage spike occurs on the pin.
R5	3.9 Ω	Current sense and load resistor. Converts the load current into a voltage feedback signal for the op-amp. Determines the load current magnitude.
U1:A	LM358	Op-amp configured as a constant-current controller. Compares the voltage across R5 to the 0.455 V reference and drives the transistor base to regulate current.
Q1	BD139	NPN power transistor acting as the variable load element. Carries the full load current from the battery to ground under control of the op-amp.
SIM1	Arduino Uno	Microcontroller. Provides the 5V supply for the op-amp and LCD, reads the battery voltage via its 10-bit ADC on pin A0, and drives the display over I2C.

LCD1	JHD 2x16 I2C	16-column, 2-row character display with I2C backpack. Shows the measured voltage, status label, and graphical health bar.
CELL1	1.5V AA	The battery under test. Connected to the circuit as the supply for the load path and the source for the voltage measurement.

4. Circuit Design and Working Principle

The circuit can be understood in three separate layers that work simultaneously: a constant-current load layer, a voltage measurement layer, and a display layer. These are described individually below, followed by an explanation of how they operate together.

4.1 Reference Voltage Divider (R1 and R2)

R1 and R2 are connected in series from Arduino's 5V output to ground. Their junction produces a stable reference voltage that is fed to the non-inverting input (pin 3) of the LM358 op-amp. This reference voltage determines the load current.

The reference voltage is calculated using the voltage divider formula:

$$\begin{aligned} V_{\text{ref}} &= 5V \times R2 / (R1 + R2) \\ V_{\text{ref}} &= 5 \times 1000 / (10000 + 1000) \\ V_{\text{ref}} &= 0.455 V \end{aligned}$$

This value was not chosen arbitrarily. The design started from a target load current of 116 mA, which is a realistic drain for small battery-powered devices. Working backwards using the load resistor R5:

$$V_{\text{ref}} = I_{\text{load}} \times R5 = 0.116 A \times 3.9 \text{ ohm} = 0.455 V$$

The divider was then designed to produce exactly that voltage. If R1 or R2 were changed, both the reference voltage and the load current would shift proportionally.

4.2 Op-Amp Constant-Current Controller (LM358)

The LM358 op-amp is the heart of the load circuit. It is wired in a negative feedback configuration to keep the load current constant regardless of the battery's voltage.

The key pins of the LM358 in this circuit are:

- **Pin 3 (non-inverting input +):** Receives the 0.455 V reference from the R1/R2 divider. This is the target the op-amp tries to maintain.
- **Pin 2 (Inverting input -):** Connected to the top of R5, which is also the emitter of the BD139. This is the feedback signal showing the actual voltage across R5.
- **Pin 1 (Output):** Drives the base of the BD139 transistor through R3. It adjusts continuously to eliminate the difference between the two inputs.
- **Pin 8 (V+):** Powered from the Arduino 5V rail.

The feedback loop works as follows. If the current through R5 is too low, the voltage at pin 2 drops below 0.455V. The op-amp output rises, turning the transistor on harder, increasing current until R5 voltage returns to 0.455V. If current is too high, the reverse happens. The loop continuously self-corrects, keeping the load at exactly 116 mA.

4.3 Transistor as the Load Element (BD139)

The BD139 is an NPN power transistor that carries the full load current from the battery to ground. The op-amp controls how much it conducts by adjusting the base drive through R3.

The current path through the transistor is: Battery positive → Collector → Emitter → R5 → Ground. The emitter must be connected to R5 specifically because that is the op-amp's feedback point. If the emitter were connected anywhere else, the feedback loop would break and current regulation would fail.

R3 between the op-amp output and the transistor base serves two purposes. It limits the base current to a safe value to protect both the transistor and the op-amp's output stage, and it introduces a small amount of damping to prevent the feedback loop from oscillating at high frequencies.

4.4 Voltage Measurement Path (R4 and Arduino ADC)

The voltage measurement path is entirely separate from the load path. R4 connects directly from the battery positive terminal to Arduino pin A0. Because the Arduino's ADC input impedance is approximately 100 MΩ, the voltage divider formed by R4 and the ADC drops only 0.1% of the measured voltage, which is completely negligible.

This separation is a deliberate design decision. If the measurement were taken from inside the load path, for example from the collector or across R5, the reading would be distorted by resistor drops and transistor saturation voltage. By measuring directly at the battery terminal through a dedicated, high-impedance path, the Arduino reads the true terminal voltage of the battery under load.

R4 also protects the ATmega328P by limiting the current that flows into pin A0 if a voltage spike or wiring mistake causes the pin to be driven above 5V or below ground.

4.5 Display System (JHD 2x16 I2C LCD)

The LCD communicates with the Arduino over the I2C protocol using only two signal wires. Pin A4 on the Arduino carries SDA (Serial Data) and pin A5 carries SCL (Serial Clock). This leaves the remaining analog and digital pins free and keeps wiring simple.

The top row of the display shows the measured voltage to three decimal places alongside the status label. The bottom row shows a graphical health bar made of 16 filled block characters scaled to the measured voltage. A fully charged cell fills all 16 blocks; a cell near the dead threshold shows only a few.

4.6 Full Circuit Operation

When a battery is inserted into the test terminals, all three layers of the circuit activate simultaneously. The reference divider puts 0.455V on the op-amp. The op-amp immediately begins driving the BD139 base, drawing current through the load path. Within a few milliseconds the current settles at 116 mA and remains there continuously. In parallel, the Arduino samples pin A0 sixty times with a 2ms delay between samples, averages the readings, converts the average to a voltage, maps it to a status label and health bar count, and updates the LCD. This cycle repeats approximately every 620 ms.

5. Key Calculations

Table 2: Key Calculations

What is Being Calculated	Formula	Result
Reference voltage (op-amp target)	$5V \times 1k / (10k + 1k)$	0.455 V
Load current drawn from battery	$0.455\text{ V} / 3.9\text{ ohm}$	approximately 116 mA
Power dissipated in R5	$(0.116)^2 \times 3.9\text{ ohm}$	approximately 52 mW (safe for 1/4 W rating)

Power dissipated in BD139	$1.5\text{V} \times 0.116\text{ A}$	approximately 174 mW (safe, rated 1.25 W)
ADC conversion to voltage	$\text{ADC_count} \times (5.0 / 1023)$	Each ADC step is approximately 4.89 mV
Display update cycle time	60 samples $\times$ 2ms + 500ms delay	approximately 620 ms per update

6. Software Implementation

The Arduino sketch is structured around four helper functions called from a main loop. Each function has a single responsibility, which keeps the code straightforward and easy to follow.

6.1 Program Structure

Table 3: Explanation of functions in arduino program

Section	Purpose
#include / #define	Imports libraries and defines all pin numbers, voltage thresholds, and ADC constants as named values.
Global declarations	Creates the LCD object and forward-declares all helper functions so the compiler knows they exist before their definitions.
setup ()	Runs once at power-on. Initializes the LCD, turns on the backlight, shows a splash screen for 2 seconds, then clears the display.
loop ()	Runs continuously. Calls readBatteryVoltage (), gets the status and bar count, and updates both rows of the LCD, then waits 500 ms.
readBatteryVoltage ()	Samples pin A0 sixty times with a 2ms delay between readings, averages the ADC values, and converts to volts.
getHealthBars ()	Maps the measured voltage linearly to a number between 0 and 16 representing the health bar fill level.
getBatteryStatus ()	Returns a plain-text status string (GOOD, OK, LOW, or DEAD) based on the measured voltage.
printHealthBar ()	Writes the health bar to the second row of the LCD using the built-in solid block character at code 255.

6.2 readBatteryVoltage ()

This function is the most important part of the code from a measurement accuracy standpoint. Taking 60 samples and averaging them reduces random electrical noise by a factor of approximately 7.7 (the square root of 60), giving a measurement accuracy of roughly plus or minus 1.3 mV. The 2ms delay between each sample allows the ADC's internal sample-and-hold capacitor enough time to fully settle through R4's 100 k Ω source resistance.

The running total is stored in a long integer rather than a regular int because 60 multiplied by the maximum ADC value of 1023 gives 61,380, which overflows a 16-bit integer. The division is done using 60.0 rather than 60 to force floating-point arithmetic and preserve the decimal precision needed for the voltage conversion.

6.3 getBatteryStatus () and Thresholds

The voltage thresholds used by getBatteryStatus () are calibrated for a battery that is under a 116-mA load, not at rest. A loaded cell has a lower terminal voltage than the same cell measured open circuit. This is why the GOOD threshold is set at 1.30V rather than 1.50V.

Table 4: Voltage threshold and corresponding message displayed on screen

Voltage Range (under load)	Status Label	Meaning
Below 0.10 V	No Cell	Nothing is connected to the test terminals.
0.10 V to 0.90 V	DEAD	The cell is fully exhausted and should be replaced.
0.90 V to 1.10 V	LOW	The cell is nearly empty and will fail soon under any real load.
1.10 V to 1.30 V	OK	The cell still has usable charge and will work in low-drain devices.
Above 1.30 V	GOOD	The cell is healthy and suitable for all normal applications.

6.4 Complete Arduino Code

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
```

```

#define LCD_ADDRESS 0x27
#define BATTERY_PIN A0
#define ADC_REF_VOLTAGE 5.00
#define ADC_MAX_VALUE 1023.0
#define NO_CELL_VOLTAGE 0.10
#define EMPTY_VOLTAGE 0.90
#define FULL_VOLTAGE 1.50

```

```

LiquidCrystal_I2C lcd (LCD_ADDRESS, 16, 2);

```

```

float readBatteryVoltage ();
int getHealthBars (float voltage);
const char* getBatteryStatus (float voltage);
void printHealthBar (int bars);

```

```

void setup () {
  lcd.init();
  lcd.backlight();
  pinMode (BATTERY_PIN, INPUT);
  lcd.setCursor(0, 0);
  lcd.print ("Battery Tester");
  lcd.setCursor(0, 1);
  lcd.print ("Loaded Voltage");
  delay (2000);
  lcd.clear();
}

```

```

void loop () {
  float voltage = readBatteryVoltage ();
  lcd.setCursor(0, 0);
  if (voltage < NO_CELL_VOLTAGE) {
    lcd.print ("No Cell ");
    lcd.setCursor(0, 1); lcd.print (" ");
  } else {
    int bars = getHealthBars(voltage);
    const char* status = getBatteryStatus(voltage);
    lcd.print("V:"); lcd.print (voltage, 3);
    lcd.print ("V "); lcd.print(status); lcd.print (" ");
    printHealthBar(bars);
  }
  delay (500);
}

```

```

float readBatteryVoltage () {
  long total = 0;
  for (int i = 0; i < 60; i++) {

```

```

    total += analogRead (BATTERY_PIN);
    delay (2);
}
float averageAdc = total / 60.0;
return averageAdc * (ADC_REF_VOLTAGE / ADC_MAX_VALUE);
}

int getHealthBars (float voltage) {
    if (voltage <= EMPTY_VOLTAGE)
        return 0;
    if (voltage >= FULL_VOLTAGE)
        return 16;
    float health = (voltage - EMPTY_VOLTAGE) / (FULL_VOLTAGE -
EMPTY_VOLTAGE);
    return (int) (health * 16.0);
}

const char* getBatteryStatus (float voltage) {
    if (voltage < 0.90) return "DEAD";
    if (voltage < 1.10) return "LOW";
    if (voltage < 1.30) return "OK";
    return "GOOD";
}

void printHealthBar (int bars) {
    lcd.setCursor(0, 1);
    for (int i = 0; i < 16; i++) {
        if (i < bars)
            lcd.write(byte (255));
        else
            lcd.print (" ");
    }
}

```

7. Results and Testing

The circuit was tested with three AA cells of varying charge levels. In each case the LCD updated within approximately 620 ms of the battery being connected and showed a stable reading that did not drift noticeably between refresh cycles. The 60-sample averaging effectively eliminated the flicker that was observed in initial testing with a single sample per cycle.

7.1 Sample Outputs

The following sample outputs illustrate the system's behavior across different battery conditions:

Fresh Cell (GOOD)

Row 0: V:1.422V GOOD

Row 1: [#####]

Partially Depleted Cell (OK)

Row 0: V:1.195V OK

Row 1: [#####]

Near-Empty Cell (LOW)

Row 0: V:0.947V LOW

Row 1: [#]

Fully Exhausted Cell (DEAD)

Row 0: V:0.612V DEAD

Row 1: []

No Cell Connected

Row 0: No Cell

Row 1: (blank)

7.2 Practical Performance

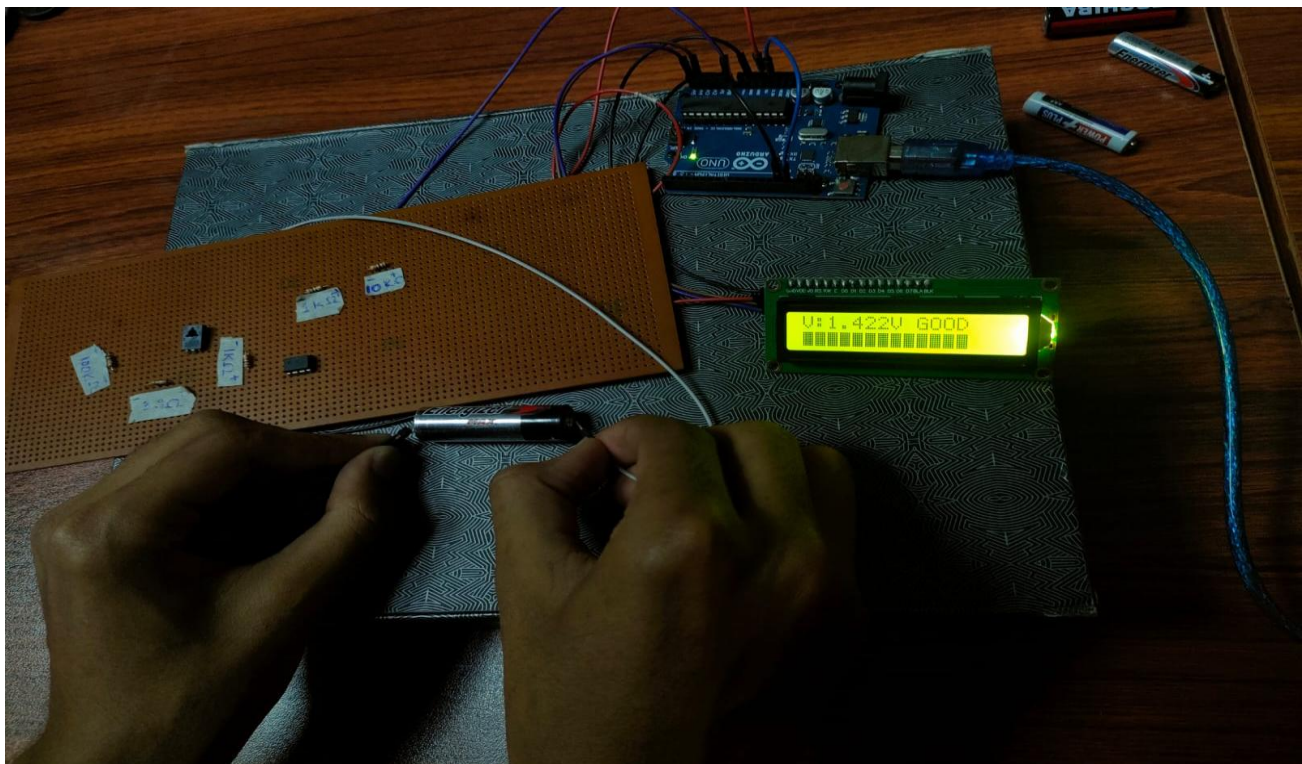


Figure 1: Constant-current load circuit during live battery testing with voltage displayed on LCD

The constant-current load circuit behaved as designed. The voltage across R5 stabilized at approximately 0.455V within a few milliseconds of battery insertion, confirming that the op-amp feedback loop was regulating correctly. The BD139 transistor remained cool throughout testing, consistent with the calculated 174 mW dissipation which is well within its 1.25W rating.

Thus, it was tried with various different cells and corresponding correct results were given, as it has been explained. Furthermore, the labelling had been done for each resistor, so as to make it easier to tell which one is positioned where.

7.3 Schematic Diagram

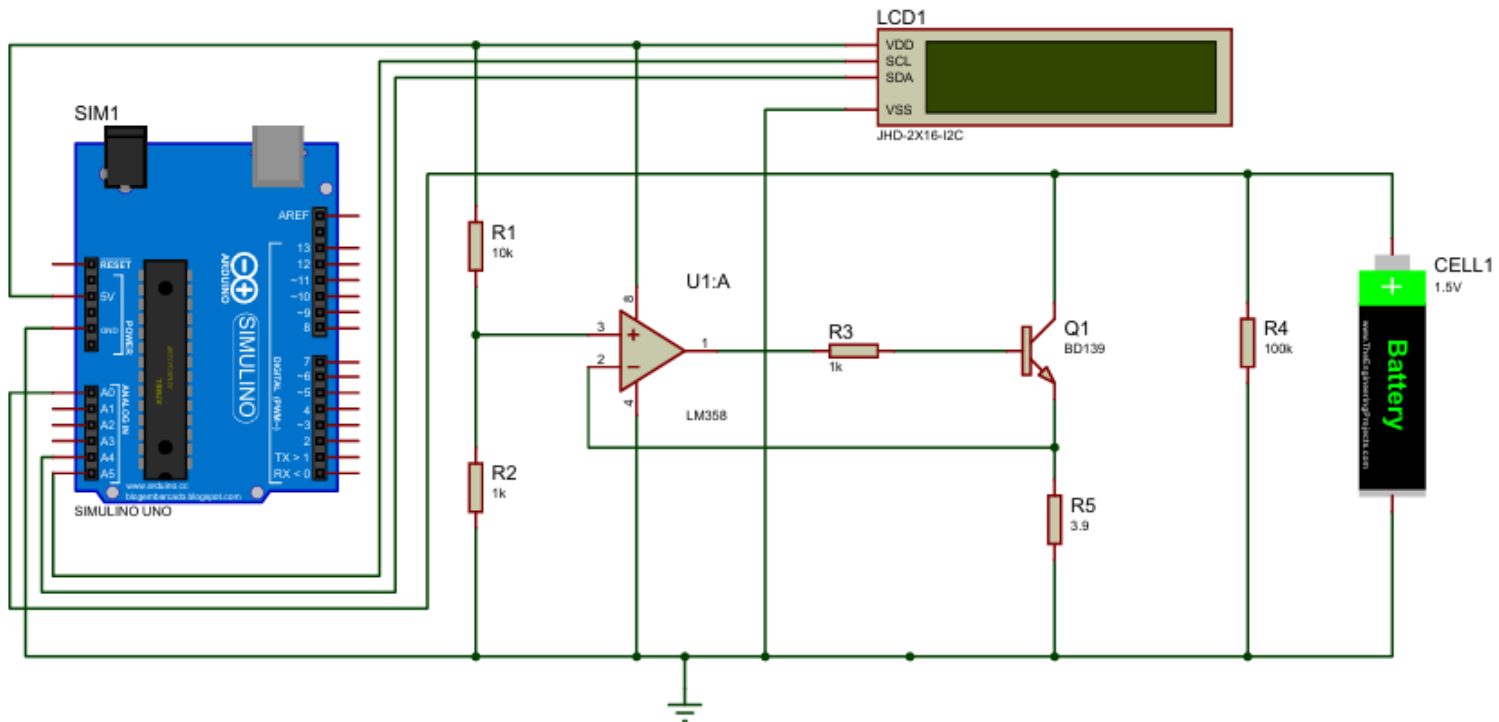


Figure 2: Schematic diagram of the constant-current load circuit showing Arduino, LM358 op-amp, BD139 transistor, and LCD voltage display

All the components, their functionality, and their corresponding roles are given an overview in the above provided Schematic Diagram.

8. Conclusion

The AA Battery Voltage Tester successfully achieved its design objective. The constant-current load circuit based on the LM358 op-amp and BD139 transistor maintained a stable 116 mA drain on the battery under test, and the Arduino ADC measured the resulting terminal voltage accurately via a separate high-impedance measurement path. All readings were displayed clearly and updated reliably on the I2C LCD.

The project brought together several core concepts from the Linear Circuit Analysis course in a single working design. The voltage divider formed by R1 and R2 was applied to set a precise reference voltage. The op-amp was used in a closed negative feedback loop to regulate current, demonstrating the practical use of the differential amplifier principle. The transistor was analyzed and sized based on its collector current and power dissipation to ensure safe operation. The ADC input protection resistor was chosen with awareness of the voltage divider it forms with the ADC input impedance.

One limitation of the current design is that all data is lost when the circuit is powered down. A possible improvement would be to log measurements to an SD card or transmit them wirelessly, enabling comparison of a battery's condition over time. Another enhancement would be to support multiple battery chemistries such as NiMH or lithium by adjusting the voltage thresholds through a hardware selector switch. Despite these potential additions, the project functions correctly as designed and meets the requirements of the semester project.

End of Report